# Multiverse, Black Hole Mechanism and Finite Core

GDS (Gravitational Dynamic System)

Martin Štěpánů
Independent Researcher
ORCID: 0009-0007-0409-6037

**Abstract**

The standard picture of general relativity leads, inside black holes, to a classical singularity: a state of infinite density and the breakdown of the physical interpretation of space and time. This result is mathematically formal, but physically problematic, because infinity itself does not represent a true explanation of nature.

In this work, we present the numerical framework of GDS (Gravitational Dynamic System), in which the singularity is replaced by a finite saturated core emerging at extreme densities. The dynamic field introduces a nonlinear saturation of the gravitational response, which stops runaway collapse and enables controlled matter transfer into a causally disconnected FRW-like expanding region.

The numerical results show: horizon crossing, formation of a finite core without singularity, exact mass conservation, causal separation without return geodesics, stable expansion of the daughter region, an early window for structure formation, the possibility of a shared daughter spacetime for multiple black holes, energetic consistency when the dynamic field is included, and robust stability under perturbations.

These results support the interpretation that black-hole interiors do not have to terminate in singularities, but may function as a mechanism of cosmological branching and the birth of new cosmic regions. Black holes may therefore represent sites of cosmogenesis rather than terminal points of physics.

## 1 Introduction

### 1.1 Singularity as a Problem of Standard Gravity

The standard solution of general relativity leads gravitational collapse to a singularity, a state in which density, curvature, and other physical quantities diverge to infinity. Such a result is mathematically possible, but physically unsatisfactory.

A singularity does not describe a mechanism, but rather the failure of the description itself. At the point of divergence, it is no longer possible to uniquely define geometry, causality, or information flow. This creates not only the singularity problem itself, but also the well-known black-hole information paradox.

From a physical point of view, it is therefore legitimate to ask whether an infinite singularity is merely a sign of model incompleteness and whether nature instead uses another dynamically regulated mechanism.

### 1.2 The Need for a Dynamic Field

The same problem also appears in cosmology. The standard ΛCDM model requires the introduction of dark matter and dark energy as effective components necessary to explain observational data. However, these components are not directly derived from fundamental dynamics.

This motivates the search for a more unified principle in which gravitational anomalies, extreme collapse, and cosmological expansion are different manifestations of a single dynamic field.

### 1.3 GDS as a Unifying Principle

In this work, we use the GDS framework (Gravitational Dynamic System), in which the gravitational response at high densities does not continue linearly, but enters a saturated regime.

The basic hypothesis is:

a black hole does not end in a singularity,
but in a finite dynamic core.

This finite core may function as a transition region enabling the formation of a causally disconnected expanding daughter-universe branch.

## 2 Theoretical Framework of GDS

### 2.1 Effective Density

At extreme densities, we introduce the effective density in the form

$$\rho_{\text{eff}} = \frac{\rho}{1 + \left(\frac{\rho}{\rho_c}\right)^n} \tag{1}$$

where $\rho_c$ is the critical saturation density and $n$ determines the sharpness of the transition.

This form ensures that at low densities the system behaves in the standard way, while at extreme densities runaway collapse is dynamically limited.

### 2.2 Dynamic Field Response

The field response is described by the function

$$\Gamma(\rho) = 1 + \frac{A}{1 + \left(\frac{\rho}{\rho_c}\right)^n} \tag{2}$$

where the parameter $A$ determines the amplitude of dynamic enhancement.

This nonlinear response modifies the effective gravitational regime and enables the formation of a stable finite core instead of divergence.

# 3 Numerical Framework

The model is divided into two regions:

- Region A: a collapsing black-hole-like region
- Region B: a daughter FRW-like expanding region

Between them there exists a saturated core with radius $R_{\text{core}}$, which functions as a dynamic interlink enabling controlled matter transfer.

The basic model parameters are:

$$\rho_c,\ n,\ A,\ R_{\text{core}},\ k_{\text{bounce}},\ \text{transfer\_strength}$$

The numerical integration was performed with high time resolution and subsequently verified using extensive scans of the parameter space.

# 4 Horizon Crossing and Finite Core

The Schwarzschild horizon was tested as

$$R_s = \frac{2GM}{c^2} \tag{3}$$

with the result

$$R_{\min} < R_s \tag{4}$$

specifically

$$R_s = 2.0, \quad R_{\min} = 0.08$$

This means that the collapse truly enters below the event horizon and is not merely stopped before the formation of a black hole.

Instead of a singularity, a finite saturated core is formed:

$$R_{\text{core}} = 0.08$$

This represents a physically finite state without infinite divergence.

## 5 Matter Transfer Mechanism

After the saturated regime is reached, a transition gate is activated:

$$\text{saturation} \to \text{gate} \to \text{flux}$$

This gate enables a controlled flow of matter from region A into region B.

The control test showed:

- transfer OFF → the daughter region does not form
- transfer ON → daughter expansion appears

This confirms the causal character of the mechanism.

At the same time, exact mass conservation holds:

$$M_A + M_B = \text{const} \tag{5}$$

with numerical accuracy at the level of machine precision.

## 6 FRW-like Daughter Region

Region B exhibits stable expansion characterized by

$$H_B = \frac{\dot{R}_B}{R_B} \tag{6}$$

with a persistently positive value of $H_B$.

This means that the result is neither a numerical artifact nor a temporary rebound, but an actual expanding spacetime-like region within the model.

The return-geodesic test further showed

$$\text{returned\_to\_core} = \text{False}$$

meaning that after entering region B, no return trajectory back into the original collapse exists.

This confirms the causal separation of the two regions.

## 7 Material Evolution and Structure Formation

The transferred matter in region B does not disappear and does not immediately break down. It forms a persistent expanding medium with long-term dynamic stability.

A comparison was analyzed between the free-fall time and the expansion time scale:

$$t_{ff} < t_H \tag{7}$$

in the early time interval approximately

$$t \approx 8.05 \to 8.51$$

This represents a window in which gravity can locally overcome expansion and enable the formation of proto-structures and proto-galaxies.

The daughter region is therefore not merely an empty expansion channel, but an environment capable of its own cosmological evolution.

# 8 Shared Multiverse Branch

Scenarios with two and later ten independent black holes were tested.

In the independent regime, different expansion rates arise:

$$H_1 \neq H_2 \neq H_3 \ldots$$

whereas in the shared regime, a single common expansion number appears:

$$H_{\text{shared}}$$

while preserving the same total transferred mass.

This supports the interpretation that multiple black holes can feed one shared daughter spacetime instead of isolated separated pockets.

This result represents numerical support for a mechanism of multiverse branching.

# 9 Energetic Consistency

The first simple test of mechanical energy showed a significant drift, which meant that mechanical energy alone is not the correct closed budget.

After introducing the energy reservoir of the dynamic field, we obtain

$$E_{\text{total}} = E_{\text{mech}} + E_{\text{field}} \tag{8}$$

The result is:

$$\text{relative total-energy drift} \sim 10^{-15}$$

which corresponds to practically exact numerical conservation of total energy.

This confirms that the daughter-universe mechanism is not a numerical trick, but an energetically consistent dynamic process.

# 10 Stability Under Perturbations

A perturbation stability test was performed with random changes of the initial conditions up to the level of several percent.

In total, 250 perturbed runs were performed, with the result

250 / 250 successful

This means that the mechanism is not a fine-tuned trajectory, but a robust dynamic attractor.

The daughter universe forms stably even under small perturbations of the system.

# 11 Discussion

The results of this work show not only an alternative to singularity, but a complete mechanism of cosmological continuation inside the black-hole regime.

The process sequence is:

black hole → horizon crossing → finite core → interlink transfer → FRW daughter universe → structure formation

Thus, the black hole changes from a terminal point of physics into a site of cosmogenesis.

At the same time, the same dynamic principle of GDS connects:

- cosmological anomalies
- the dark-sector problem
- stability of nonlinear control
- the internal structure of black holes
- the mechanism of multiverse branching

This unifying capability represents a strong argument for the fundamental importance of the dynamic field.

# 12 Observable Predictions and Semiclassical Consistency

A physically relevant model must not only remain internally consistent, but must also provide observable signatures that distinguish it from the standard singular black-hole picture.

The GDS framework provides four major externally testable consequences: a gravitational-wave echo prediction, consistency with early structure formation observed by JWST, compatibility with Hawking–Bekenstein entropy, and a finite remnant scenario for Hawking evaporation.

## 12.1 LIGO Ringdown Echo Prediction

If the interior of a black hole does not contain a true singularity, but instead contains a finite saturated core, post-merger gravitational waves may partially reflect from this internal structure.

This produces a delayed echo signal after the standard ringdown phase.

For the Schwarzschild radius

$$R_s = \frac{2GM}{c^2} \tag{9}$$

and the finite GDS core radius $R_{\rm core}$, the echo delay can be estimated as

$$\Delta t_{\rm echo} \approx \frac{2R_s}{c} \ln\left(\frac{R_s}{R_{\rm core}}\right). \tag{10}$$

In the normalized model,

$$R_{\rm core} = 0.08, \tag{11}$$

with the ratio $R_s/R_{\rm core}$ approximately equal to 25.
For a remnant black hole with mass

$$M \approx 30\, M_\odot, \tag{12}$$

the predicted delay is

$$\Delta t_{\rm echo} \approx 1.9 \text{ ms}. \tag{13}$$

For the mass range $10\, M_\odot$ to $150\, M_\odot$, the model predicts echo delays from approximately

$$0.6 \text{ ms} \quad \text{to} \quad 9.5 \text{ ms}. \tag{14}$$

This interval lies directly in the range that can be tested in LIGO/Virgo/KAGRA data through post-ringdown residual analysis.

A true singularity does not provide a natural reflection mechanism of this type. The echo signal is therefore one of the most important falsifiable predictions of the GDS finite-core scenario.

## 12.2 Consistency with JWST Early Structure Formation

The daughter region in GDS exhibits an early window for structure formation defined by the condition

$$t_{ff} < t_H, \tag{15}$$

where $t_{ff}$ is the free-fall time of local gravitational collapse and $t_H$ is the expansion time scale.

This window was numerically found in the interval

$$t \approx 8.05 \to 8.51 \tag{16}$$

within the normalized daughter-region evolution.

After calibration to the early universe, this window maps approximately to the high-redshift range

$$z \approx 10.8 \to 11.3. \tag{17}$$

This range lies directly in the regime where JWST observes surprisingly early and massive galaxies.

GDS therefore provides a natural mechanism for accelerated early structure formation: during the early phase of daughter expansion, local gravity can overcome global expansion and form proto-structure and proto-galaxy seeds.

This result is not yet a final cosmological fit, but it provides an important first quantitative contact with the JWST-relevant high-redshift regime.

## 12.3 Compatibility with Hawking–Bekenstein Entropy

An important semiclassical test is whether a finite core inside a black hole violates the standard horizon entropy picture.

Hawking–Bekenstein entropy is determined by the horizon area,

$$S_{\mathrm{BH}} = \frac{A_{\mathrm{horizon}}}{4G\hbar}. \tag{18}$$

In the GDS scenario, the singularity is replaced by a finite core, but this core remains deep inside the horizon.

Numerically, the ratio of the core area to the horizon area is

$$\frac{A_{\mathrm{core}}}{A_{\mathrm{horizon}}} = 1.6 \times 10^{-3}. \tag{19}$$

The dominant entropy therefore remains associated with the horizon, not with the core.

This means that the GDS finite-core scenario is not in direct conflict with standard Hawking–Bekenstein entropy scaling. The finite core replaces the singularity, but does not remove the semiclassical dominance of the horizon area.

## 12.4 Finite Remnant Endpoint of Hawking Evaporation

Standard Hawking evaporation formally drives black holes toward very small mass and an unclear singular endpoint.

In GDS, a natural finite threshold appears when the Schwarzschild radius approaches the finite core scale,

$$R_s \to R_{\mathrm{core}}. \tag{20}$$

This corresponds to the remnant mass

$$M_{\mathrm{remnant}} = \frac{R_{\mathrm{core}} c^2}{2G}. \tag{21}$$

In the normalized units of the model,

$$M_{\mathrm{remnant}} = 0.04. \tag{22}$$

The Hawking temperature increases as expected, but evaporation does not need to continue to zero mass. Instead of a singular endpoint, a finite-core remnant endpoint emerges.

This scenario is naturally compatible with interpretations in which information is not destroyed by a singularity, but remains connected to a finite dynamic structure.

## 12.5 Summary of Observable Consequences

The extended tests show that GDS is not only an internally consistent numerical model replacing the singularity by a finite core. It also provides falsifiable predictions:

- millisecond gravitational echo signals after black-hole mergers,
- an early structure-formation window around $z \approx 10.8 \to 11.3$,
- compatibility with Hawking–Bekenstein horizon entropy,
- a finite remnant scenario for Hawking evaporation.

Thus, GDS moves from a purely conceptual finite-core model toward a physical framework that can be tested using gravitational waves, high-redshift cosmology, and semiclassical black-hole thermodynamics.

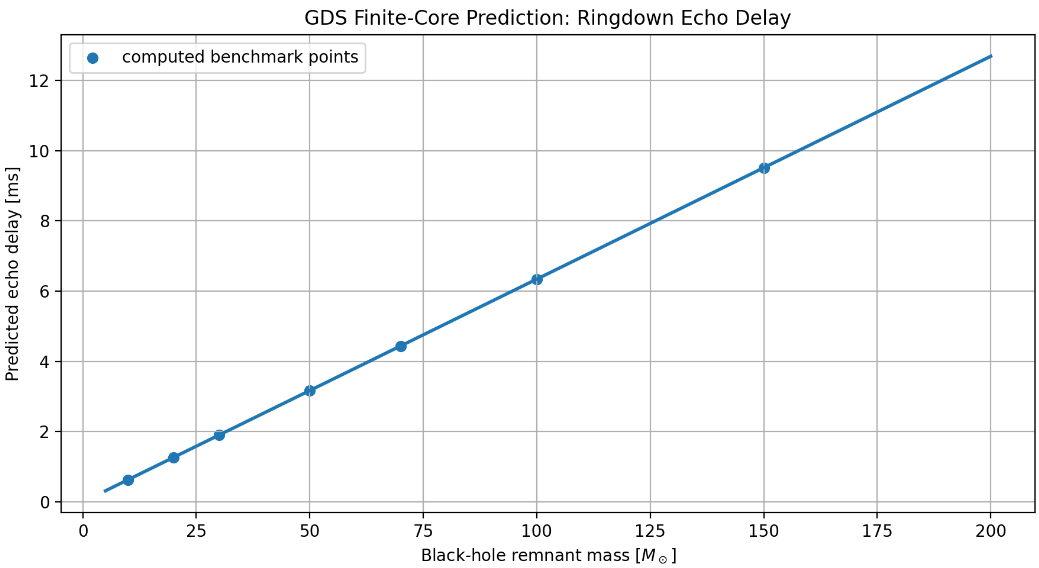


Figure 1: GDS finite-core prediction for post-ringdown gravitational-wave echo delay as a function of black-hole remnant mass.

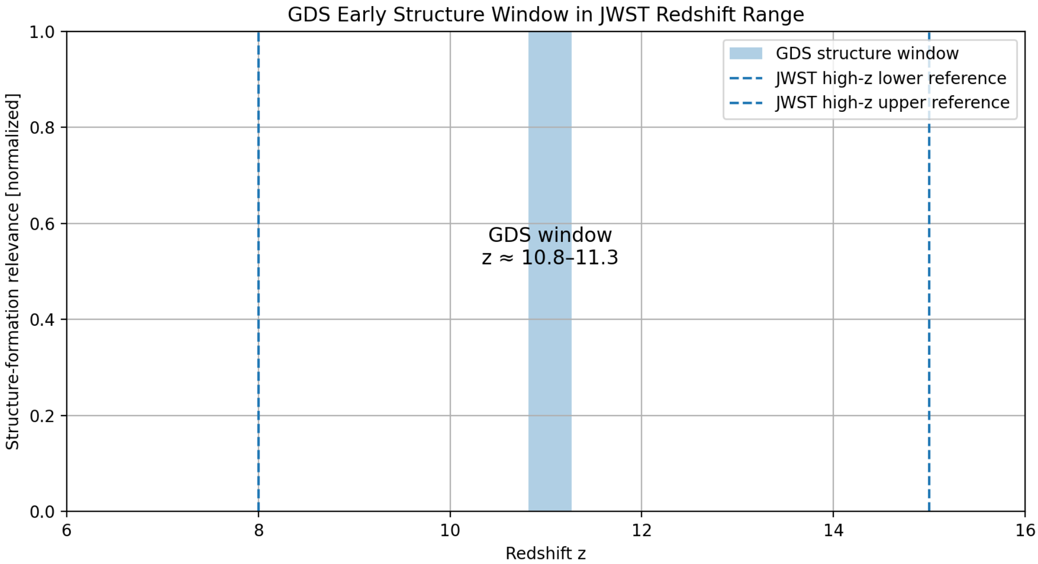


Figure 2: GDS early structure-formation window mapped into the JWST high-redshift regime.

# 13 Conclusion

The numerical results of this work support the conclusion that black-hole interiors do not have to terminate in infinite singularities.

Instead, a finite saturated core is formed, which enables controlled matter transfer into a causally disconnected FRW-like expanding region.

This region exhibits:

- stable expansion
- conservation of mass and energy
- impossibility of return
- possibility of early structure formation
- robust stability under perturbations

The results support the interpretation that black holes may function as cosmological birth sites of new universe branches.

Singularity is therefore not the final answer, but only the boundary of the old description.

GDS shows a path toward new physics in which the dynamic field replaces infinities with a real mechanism of reality.

# 14 List of Key Graphs and Figures

The graphs are stored in the `MUL/` folder. For compilation in Overleaf, this folder must be uploaded together with the `main.tex` file. The file names below exactly match the generated image files.

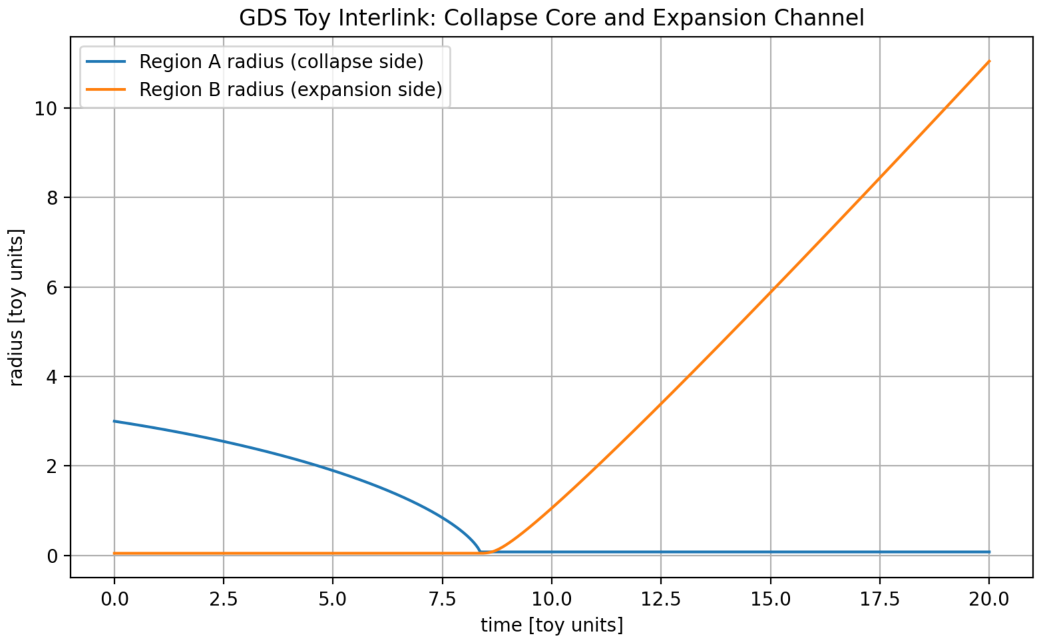


Figure 3: Evolution of the radius of collapsing region A and expanding region B. The graph shows horizon crossing, finite-core formation, and the subsequent expansion of the daughter region.

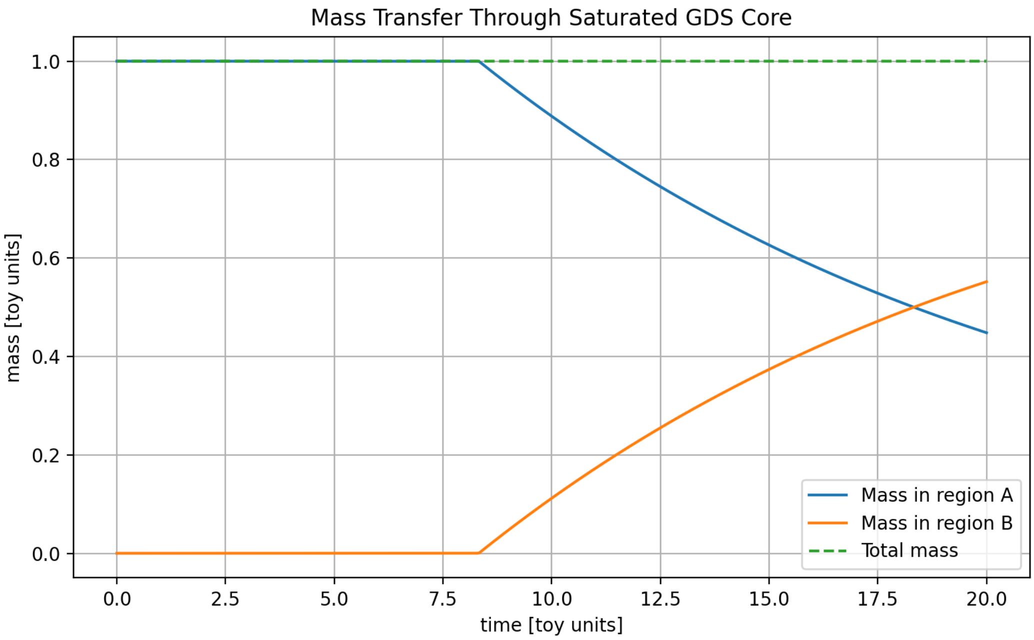


Figure 4: Matter transfer between region A and region B under exact conservation of the total mass of the system.

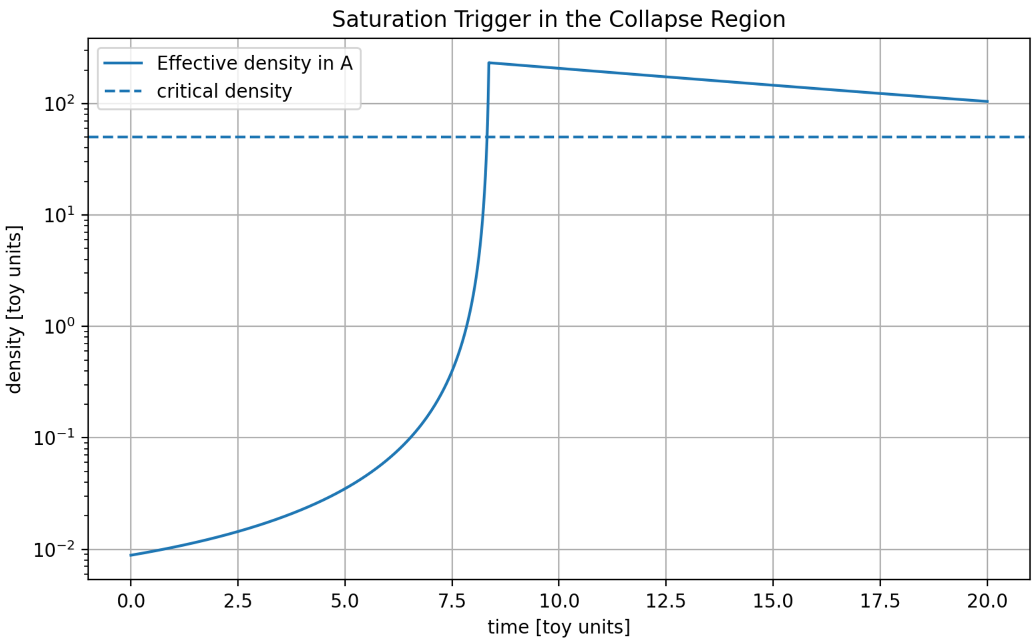


Figure 5: Growth of the effective density in collapsing region A and the reaching of the critical density that activates the saturated regime.

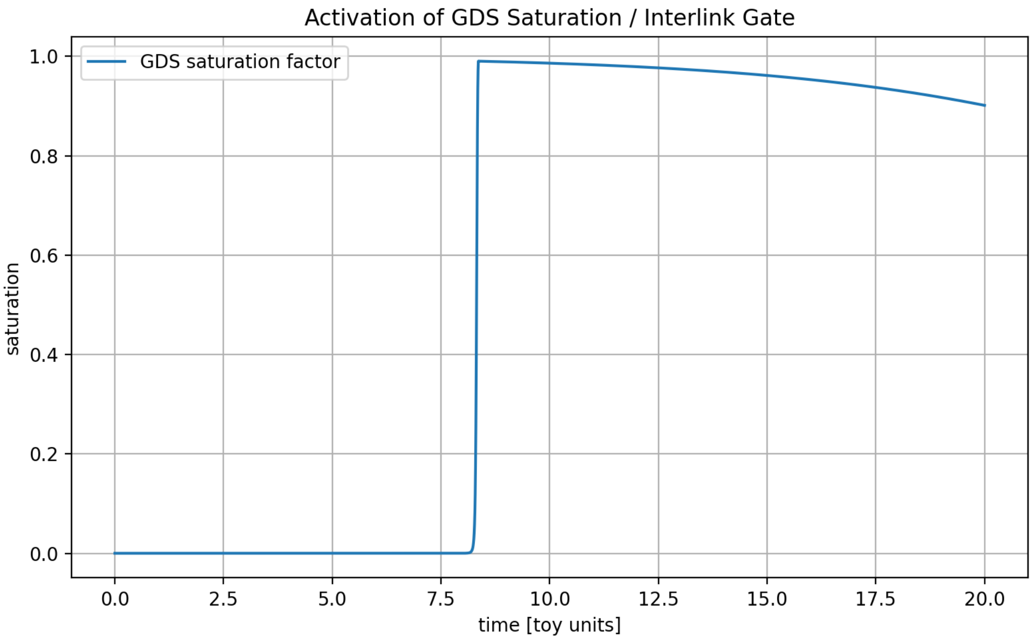


Figure 6: Activation of GDS saturation and the interlink mechanism. The graph shows the nonlinear transition from collapse to the opening of the transfer channel.

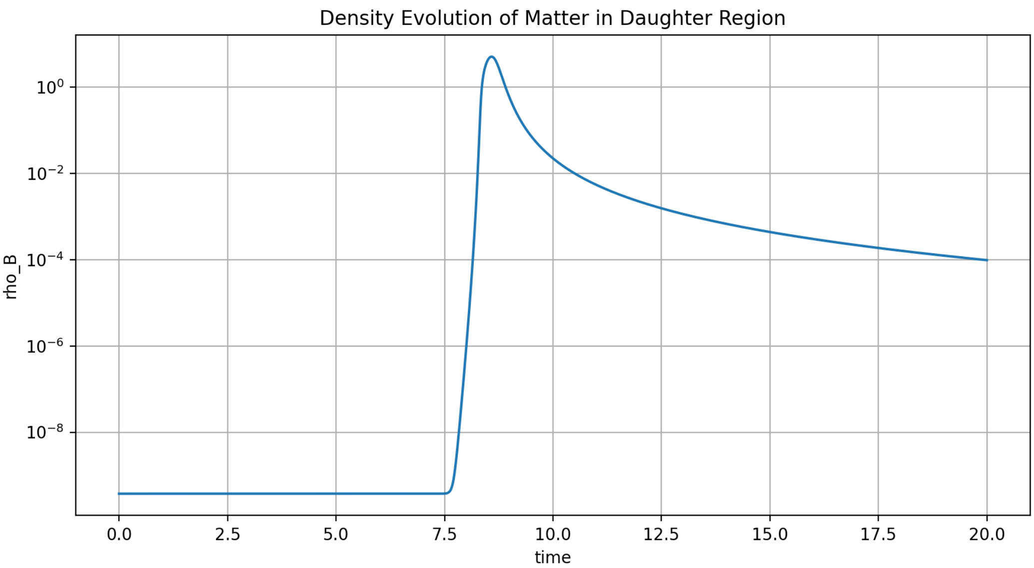


Figure 7: Material evolution in daughter region B. The transferred matter does not disappear, but dilutes into a persistent expanding medium.

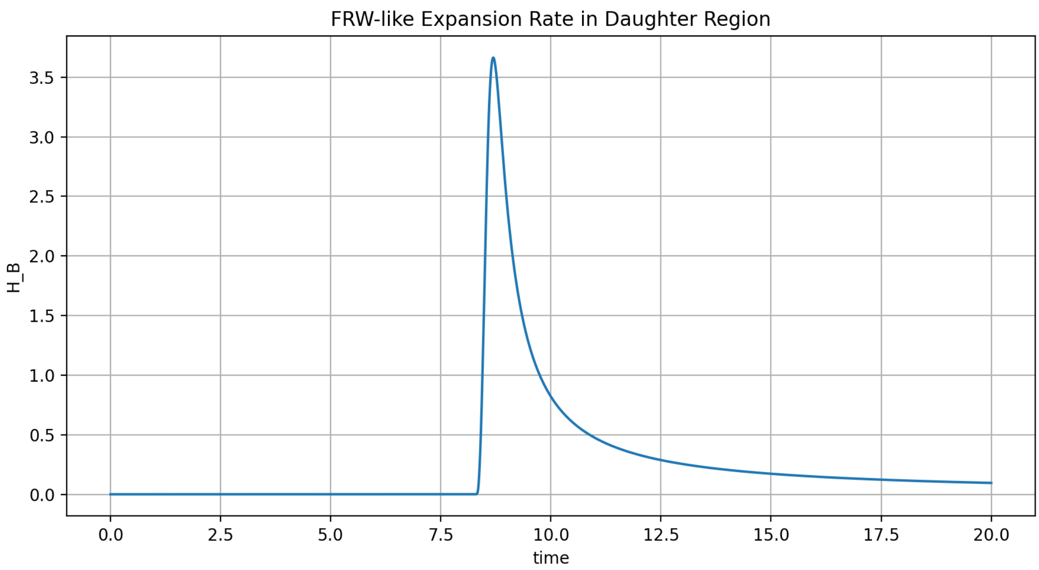


Figure 8: FRW-like expansion rate $H_B$ in the daughter region. A positive long-term value confirms stable expansion.

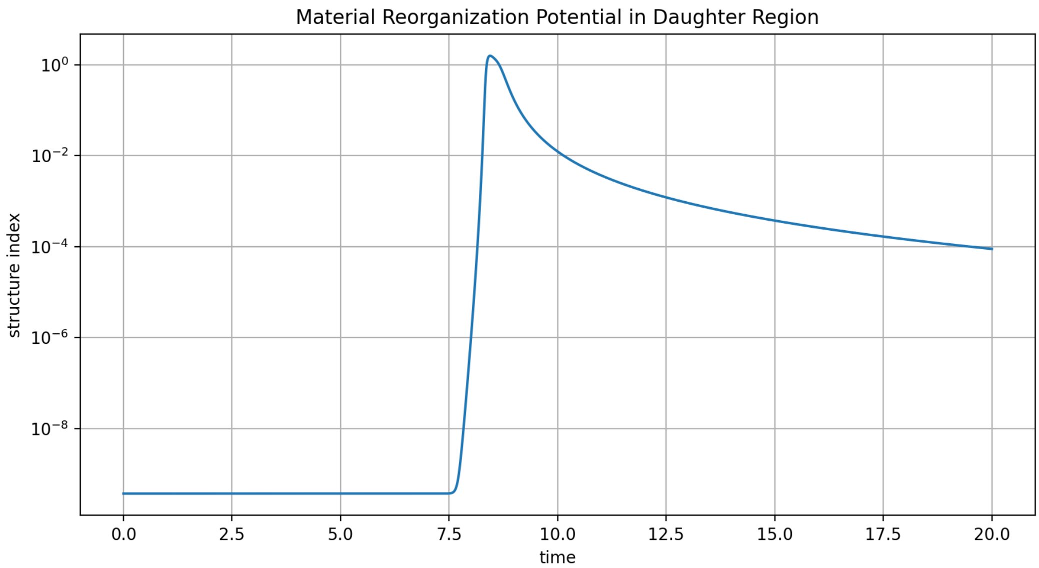


Figure 9: Index of material reorganization possibility in daughter region B. The graph documents that the transferred material can form a dynamically evolving medium.

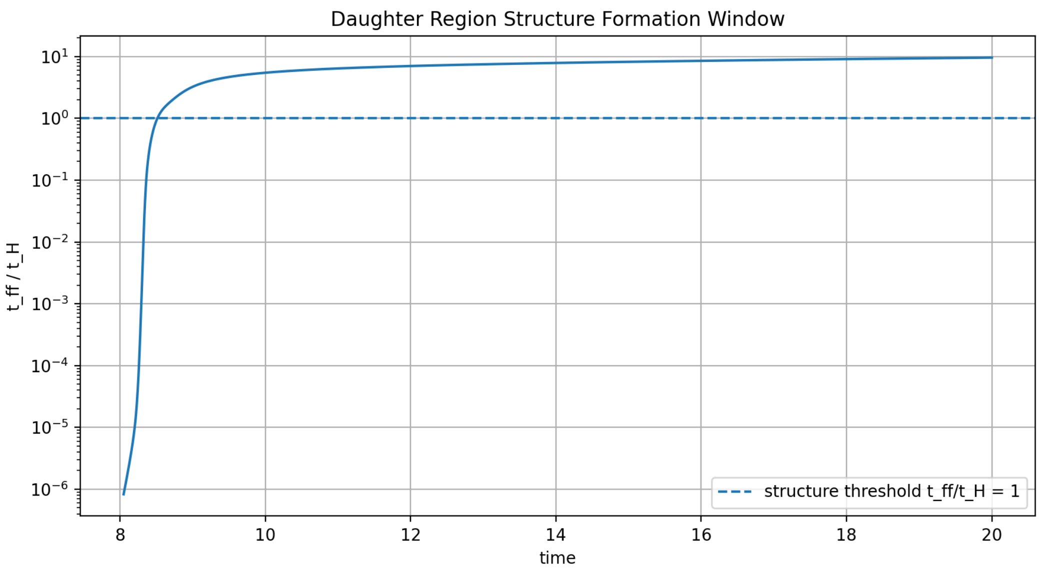


Figure 10: Window for structure formation defined by the ratio $t_{ff}/t_H$. The region below 1 corresponds to the regime in which local gravitational collapse can overcome expansion.

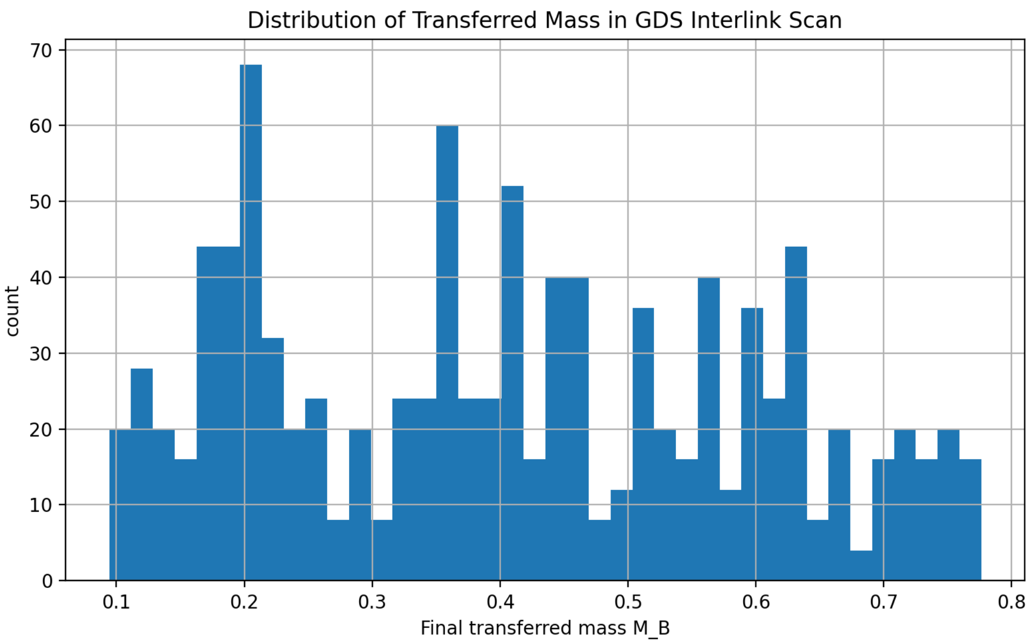


Figure 11: Histogram of transferred mass $M_B$ in the parameter scan. The result shows robust occurrence of matter transfer across the tested parameter space.

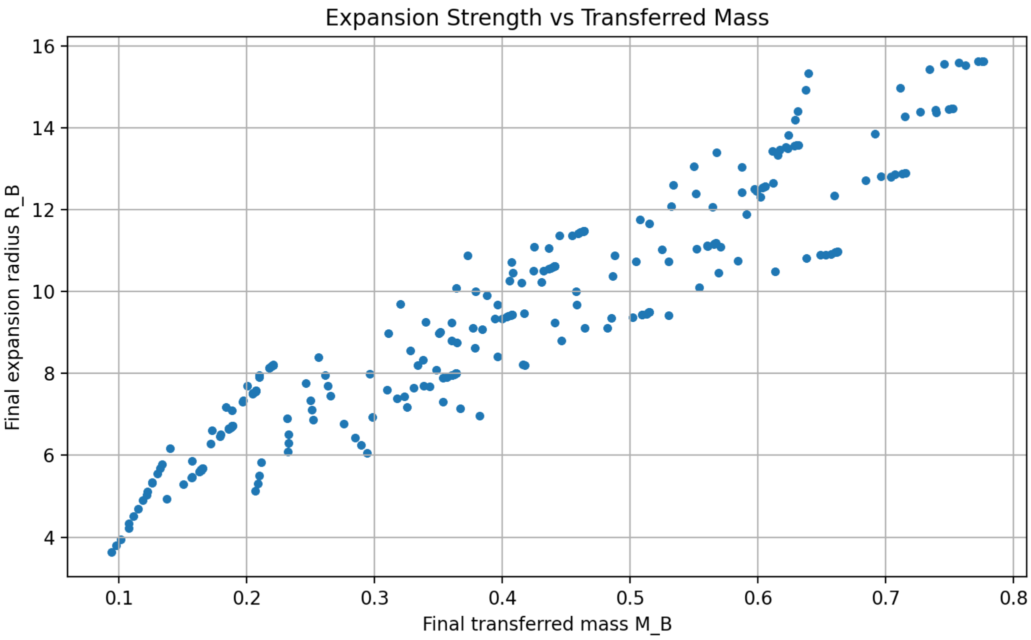


Figure 12: Dependence of the final expansion size of region B on the transferred mass $M_B$. The graph confirms the relation between interlink transfer and daughter-region expansion.

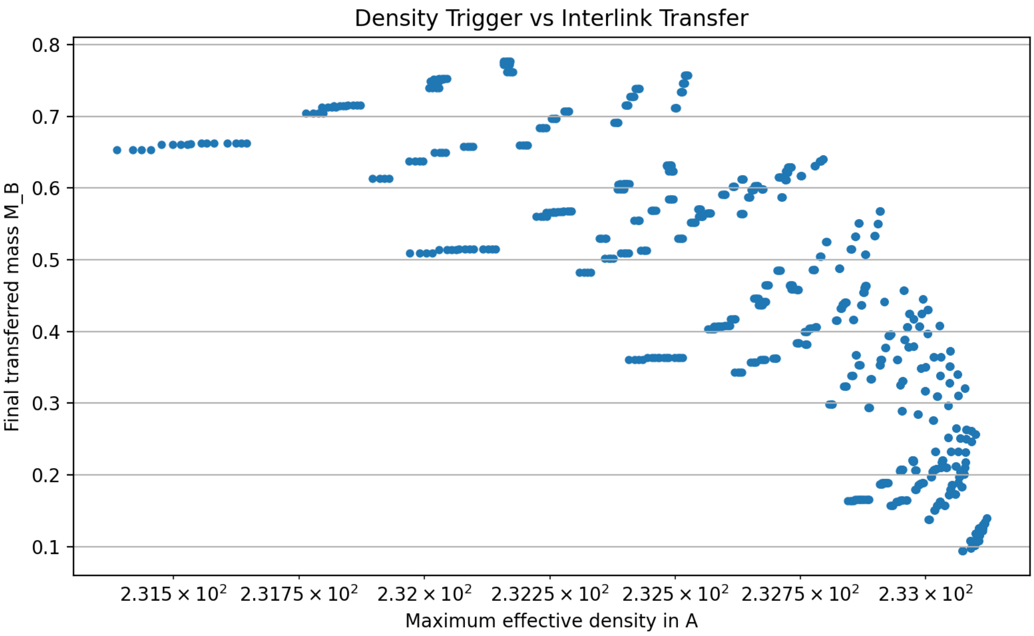


Figure 13: Relation between maximum effective density in region A and the transferred mass into region B. The graph documents the role of the density trigger in the interlink mechanism.

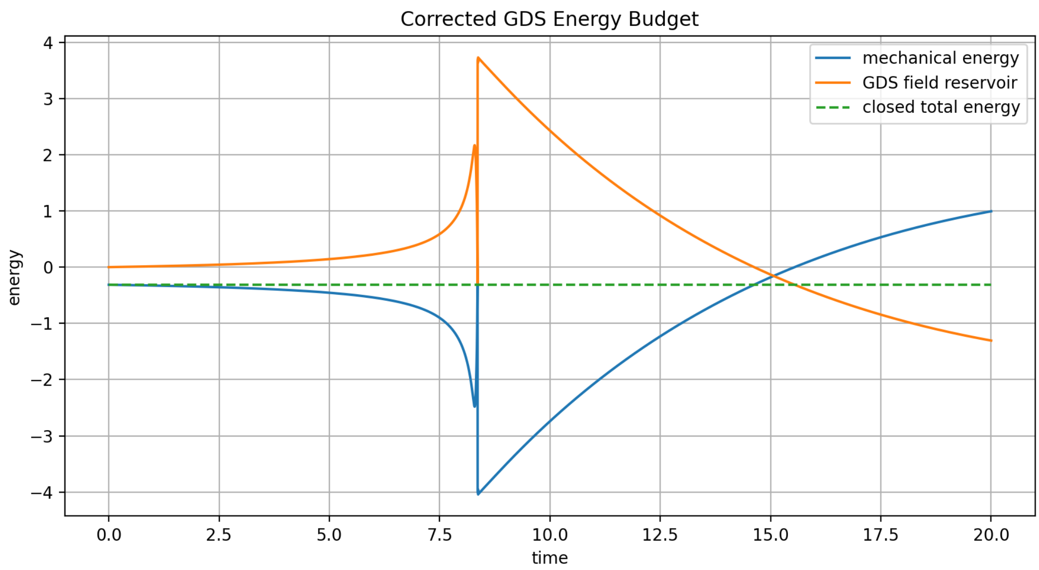


Figure 14: Corrected energy budget of the system. The total energy including the dynamic GDS field remains conserved with numerical precision of order $10^{-15}$.

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## 15 Reference

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